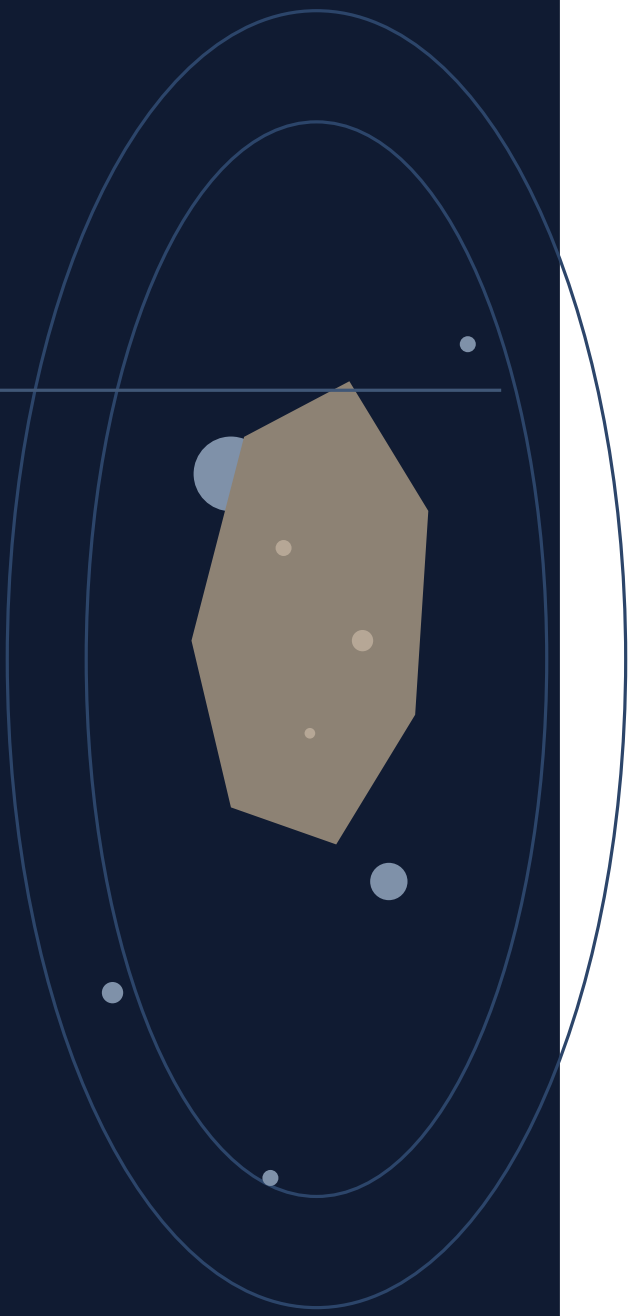
STAC WARS • PS 4

MINING THE ASTEROID BELT

A modular, water-first architecture for economically viable space resources



TEAM THE STACK AVENGERS

Team Lead: Yashica Singla

CONCEPT DESIGN REPORT | 24 OCTOBER 2026

A decision-ready concept report aligned to all three PS 4 scoring areas: engineering, cost and business.

Executive Summary

Our proposal: build a reusable, modular asteroid-resource service that mines **water ice first**, converts it into purified water and later propellant, and sells it to customers operating in cislunar space. Instead of attempting to bring bulk asteroid material to Earth, The Stack Avengers targets the nearer-term market: water delivered to orbital depots, lunar logistics operators and spacecraft that need life-support reserves or propellant feedstock.

The central economic insight is that **transporting mass out of Earth's gravity well is expensive**. If water can be sourced and processed in space, customers may avoid launching an equivalent amount from Earth. The business therefore sells a delivered service—verified water at a defined orbital location—not simply raw asteroid ore.

We propose a staged mission: robotic prospecting; a low-mass solar-electric transfer tug; a controlled anchoring and enclosed excavation system; solar-thermal extraction; purification and storage; then reusable cargo transport to a customer depot. The first mission is a technology demonstrator and resource-validation campaign, not a promise of immediate large-scale profitability.

The design deliberately limits early risk: select a target only after remote and in-situ characterization, process a small batch before scaling, use standardized replaceable modules, and secure customer demand before committing to expensive extraction hardware.

Our investment case in one sentence: prove a repeatable water-delivery service at small scale, then scale throughput only when verified demand and measured operating costs justify it.

What makes the concept credible

- **Engineering:** modular spacecraft, autonomous rendezvous, surface anchoring, enclosed thermal extraction, water purification and reusable logistics.
- **Cost:** water-first targeting, solar-electric propulsion, in-situ resource utilization, modular repair and pre-sold deliveries.
- **Business:** anchor customers in cislunar space, contract-based delivery, quality assurance and phased capacity expansion.

Important scope note

The PS asks for a technically plausible concept and proper assumptions, not a fully engineered flight design. Numerical values in this report are planning assumptions or decision metrics—not verified mission quotations, asteroid resource measurements, or guaranteed market prices. A real venture would require orbital analysis, target-specific prospecting, detailed engineering, regulatory review and customer contracts.

1. Problem Framing and Strategic Choice

Asteroid mining is not automatically profitable because the value of an extracted material must exceed the full cost of reaching the target, operating safely, processing the material, transporting it and delivering it to a paying customer. A high concentration of valuable metal does not help if the material is costly to recover or there is no practical buyer.

We compare two broad strategies:

Strategy A — Return precious metals to Earth. This may appear attractive because platinum-group metals have high terrestrial value, but it creates difficult questions around extraction, refining, re-entry, insurance, market price impact and the cost of returning large masses.

Strategy B — Use water in space. Water is useful for life support and can be split into hydrogen and oxygen propellant if suitable power and processing systems are available. Supplying water to spacecraft and depots can avoid transporting all that water from Earth. This creates a nearer-term, location-specific customer proposition.

We select Strategy B as the first commercial wedge. It avoids betting the company on a speculative return-to-Earth metal market and lets the venture scale by serving a defined logistics network. Metal extraction can be a later product line after the company proves its transport, anchoring and processing systems.

Target selection criteria

- Evidence of accessible water-bearing material, supported by remote sensing and close-range measurements.
- A trajectory and operating environment compatible with a low-energy transfer and safe rendezvous.
- Manageable spin, surface roughness, thermal conditions and communication geometry.
- A realistic delivery route to a customer depot; proximity to a paying customer matters as much as resource abundance.
- A target that allows a small pilot before the company commits to industrial-scale equipment.

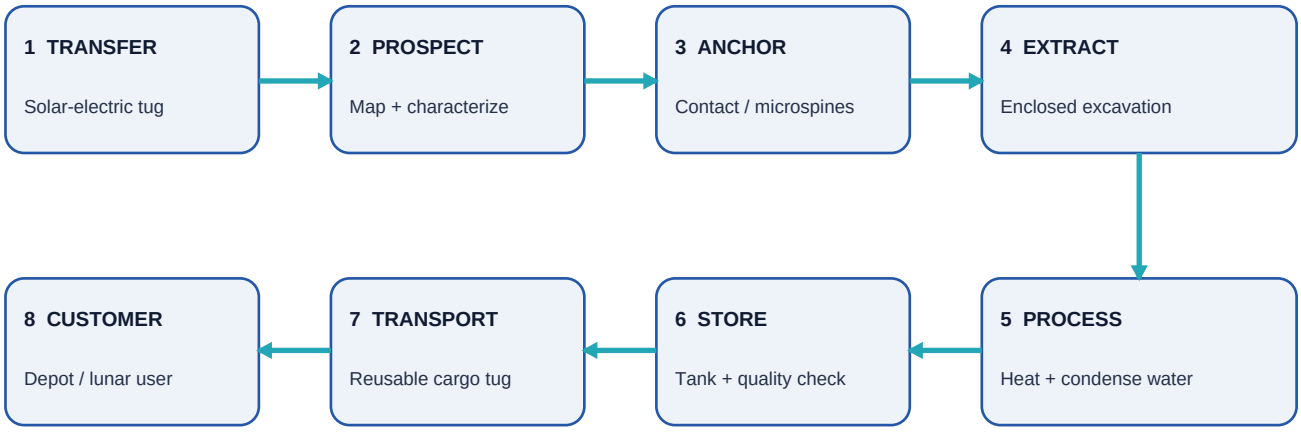
Operating principle

Do not choose an asteroid solely because it is rich in a resource. Choose the asteroid-resource-customer combination with the best total delivered-cost case. The resource is only commercially useful when it can be extracted, processed, stored and delivered at the required quality and location.

2. Engineering Architecture

The architecture is a chain of interoperable modules. Each module has a defined interface so that the company can replace or upgrade one subsystem without redesigning the entire spacecraft.

END-TO-END RESOURCE FLOW



Control layer: autonomous navigation • fault detection • Earth/relay communications

2.1 Mission phases

Phase 1 — Discover and qualify. Use telescopic observations and a prospecting spacecraft to estimate composition, shape, spin, surface properties and thermal behaviour. A close-range instrument package can include visible/near-infrared imaging, thermal sensing, and radar or other suitable subsurface-probing methods. The system must confirm water-bearing material before committing the mining vehicle.

Phase 2 — Transfer and rendezvous. A reusable solar-electric tug provides efficient long-duration transport and fine trajectory correction. Solar-electric propulsion is low-thrust, so transfer time and target geometry must be planned carefully. Chemical thrusters can be retained for time-critical manoeuvres and contingency control.

Phase 3 — Stabilize and anchor. The mining craft approaches at low relative speed, maps local hazards, and uses a contact/anchoring system selected for the target's measured surface. Candidate mechanisms include harpoons, microspine grippers, screw anchors or a reaction-force-balanced contact frame. No single anchor should be assumed to work on every asteroid; the design must be validated against local regolith and gravity conditions.

Phase 4 — Enclosed extraction. A sealed excavation hood or flexible containment enclosure limits material escape in microgravity. A low-disturbance auger, bucket-wheel or heated probe moves material into a contained processing chamber. The selected tool depends on whether the deposit is loose regolith, fractured rock or a consolidated ice-bearing layer.

Phase 5 — Thermal processing and purification. Apply controlled heat to release volatiles from excavated material. Capture vapour in a closed duct, cool it in a condenser, remove particulates and unwanted contaminants, then verify purity with onboard sensors. The system should be designed to retain and recycle heat where practical.

Phase 6 — Storage and delivery. Store purified water in insulated tanks with temperature and pressure monitoring. A reusable cargo tug transports sealed tanks or standardized tank modules to a depot. The receiving customer verifies mass, purity and delivery location before acceptance.

2.2 Energy system

Deployable solar arrays are the baseline power source. They supply the prospecting instruments, avionics, communications, thermal controls and processing equipment when illumination and geometry permit. Energy storage buffers short periods of shadow or peak loads. The thermal extraction system is the major energy consumer; therefore, the mission must measure the energy required per kilogram of recovered water before scaling. If a target has poor

illumination, the team must assess larger arrays, longer processing windows or a different target rather than assuming unlimited solar power.

2.3 Navigation, communication and autonomy

- **Navigation:** star trackers, inertial measurement units, optical navigation cameras and ranging sensors support approach, mapping and station-keeping.
- **Communications:** high-gain antenna for long-range links, low-rate safe-mode beacon, and relay support where direct Earth visibility is unavailable.
- **Autonomy:** onboard hazard mapping, relative navigation, contact monitoring, fault detection and safe-mode logic reduce dependence on continuous Earth control.
- **Operations:** Earth operators approve high-consequence actions; routine mapping, thermal cycles and health checks can run autonomously with telemetry review.

2.4 Failure handling and maintainability

The spacecraft uses redundant critical sensors, isolated power/data buses, replaceable processing modules, mechanical latches and a safe mode that stops excavation if anchoring, temperature, pressure or attitude leaves its permitted envelope. A fault tree and failure-mode-and-effects analysis should be completed before flight. The system should fail safe: stop material movement, close valves, isolate the affected module, preserve communications and wait for a verified recovery command.

3. Engineering Bottlenecks and Breakthroughs

The most difficult bottleneck is not simply digging. It is controlling the spacecraft and material while applying forces in an extremely low-gravity environment. Excavation can push the vehicle away, disturb the surface, or cause material to escape. The system must anchor itself, contain dust and vapour, and keep its thermal process stable.

Our proposed breakthrough is a **closed-loop, force-balanced mining pod**: an anchored frame with multiple independently controlled contact points, a sealed excavation hood, and sensors that continuously measure reaction forces, vibration, temperature and material flow. The controller adjusts tool speed and contact force to avoid destabilizing the vehicle. This is a development target, not a currently demonstrated off-the-shelf asteroid-mining product.

Additional high-value technology advances

- **Autonomous site selection:** combine imagery, spectral data and terrain maps to rank safe excavation zones.
- **Low-loss thermal extraction:** improve insulation, heat recovery and condenser efficiency to reduce energy per kilogram.
- **Dust-tolerant mechanisms:** sealed bearings, replaceable wear surfaces and non-contact sensing reduce maintenance.
- **Standardized cargo modules:** common tank interfaces make the tug, mining craft and depot interoperable.

Validation gates before scaling

Gate	Evidence required	Go / no-go decision
G1: Target qualification	Composition, accessible deposit, spin, surface and trajectory data	Do not launch mining hardware without a credible resource case
G2: Contact demonstration	Successful approach, anchoring and controlled tool operation in representative tests	Redesign anchoring if contact forces cannot be controlled
G3: First water batch	Measured recovered mass, purity, energy use and loss rate	Proceed only if recovery is repeatable
G4: Delivery demonstration	Tank integrity, transfer, depot interface and customer acceptance	Scale only after end-to-end delivery works

The key proof point is measured delivered cost—not a rendering, a theoretical ore value or a successful landing alone.

4. Cost Strategy: Design for Minimum Cost

The cost model must include the whole value chain: prospecting, launch and transfer, spacecraft development, operations, extraction, processing, storage, transport, insurance, ground support and customer delivery. Early missions will be expensive because they carry development and validation costs. The objective is to reduce recurring cost per accepted kilogram as the system is reused and throughput increases.

Three largest cost drivers and our response

Cost driver	Why it is expensive	Our cost-reduction design	Metric to track
1. Access and transport	Launch mass, propulsion, long mission duration, trajectory constraints and propellant	Solar-electric tug; lightweight modular payload; target chosen for total mission delta-v and customer access; reusable tug where feasible	Cost per kg delivered; propellant used; transfer time
2. Mining and processing hardware	Anchoring, excavation, thermal systems, power, dust control and reliability in microgravity	Water-first process; small pilot; modular replaceable units; shared bus; heat recovery; avoid unnecessary refining	Energy per kg water; uptime; recovery rate; maintenance hours
3. Operations and delivery risk	Long-distance control, communications, failures, schedule delays and uncertain customer demand	Autonomous routine operations; safe-mode design; standardized tanks; anchor-customer contracts; staged investment	Cost of downtime; delivery success; contracted demand

4.1 Assumptions for an early-stage economic model

Because the PS provides no asteroid, trajectory, payload, launch-price, customer-price or production data, we do not claim a precise profit forecast. Instead, the first mission establishes the variables needed for a defensible model:

- Recovered water per operating day and total recovered mass.
- Energy consumed per kilogram of accepted water.
- Mission duration, usable operating time and equipment uptime.
- Mass of mining, power, storage and transport hardware.
- Losses during extraction, purification, storage and transfer.
- Cost of launch, tug operation, communications, ground control, insurance and replacement modules.
- Customer willingness to pay for water delivered at a specific depot, including quality and reliability requirements.

4.2 Conventional approach comparison

Dimension	Conventional one-off mission	The Stack Avengers concept
Transport	Mission-specific spacecraft and propulsion; limited reuse	Reusable tug and standardized cargo interfaces where technically practical
Mining	Large integrated machine committed before resource is fully validated	Prospecting first; pilot-scale modular extraction; scale after measured results
Processing	Potentially separate, bespoke systems	Enclosed thermal extraction, condensation and purification in one process chain
Market	May focus on returning high-value material to Earth	Water delivered to in-space users; avoid unnecessary Earth return
Risk	Large upfront commitment and limited learning before major spend	Stage-gated investment; each mission validates a specific capability

Cost discipline: no scale-up until the team has measured extraction yield, energy use, system uptime and the customer's accepted delivered price. A low-cost concept on paper is not yet a low-cost

operation.

5. Business Model: Who Pays for It?

5.1 Product and value proposition

Primary product: purified water delivered in sealed, traceable tank modules to a cislunar or orbital depot. Water can support life-support systems, contingency reserves and, after suitable electrolysis and cryogenic handling, propellant production. The initial product is water; the company should not claim it can supply propellant until the additional processing and storage chain is demonstrated.

Why water? It is useful to human and robotic operations, can be stored and transferred, and may reduce the need to launch equivalent mass from Earth. The value is strongly location-dependent: water delivered where a spacecraft needs it is more valuable than the same water left on a distant asteroid.

5.2 Customer segments

- **Government space agencies:** lunar exploration, surface operations and logistics programmes.
- **Commercial orbital-depot operators:** companies developing refuelling, storage and servicing infrastructure.
- **Lunar transportation and habitat operators:** missions requiring water for crew support, shielding or propellant feedstock.
- **Spacecraft operators:** customers who may purchase contingency water or logistics capacity when the delivery point and quality are suitable.

These are target customer categories, not confirmed buyers. Before a commercial mining mission, the company must validate demand through interviews, letters of intent, pilot agreements or binding offtake contracts.

5.3 Where should the resource be sold?

Preferred: use or sell in space. Deliver water to an orbital depot or lunar logistics node selected according to customer demand and transfer economics. Returning bulk water to Earth is not the baseline because terrestrial water is widely available and re-entry/return logistics add cost without a clear value premium. Selling raw material at the asteroid is also unattractive if no customer infrastructure exists there.

5.4 Revenue model

- **Offtake contract:** customer commits to purchase a specified quantity after quality and delivery conditions are met.
- **Milestone payments:** staged payments for target qualification, successful extraction demonstration and accepted delivery reduce financing risk.
- **Delivery-as-a-service:** charge for verified kilograms delivered to a named depot, with service-level terms for purity, storage integrity and delivery window.
- **Capacity reservation:** customers pay to reserve a future delivery slot once the system has demonstrated reliable operations.

The price should be negotiated against the customer's alternative cost of obtaining water at the same location, adjusted for purity, reliability, delivery timing and contract risk. We do not assign a market price because the PS gives no evidence of a current, established asteroid-water market.

5.5 Business flywheel

Prospecting data lowers resource uncertainty → pilot extraction proves recovery and energy performance → verified delivery creates customer confidence → offtake contracts improve financing → repeat missions improve utilization and learning → standardized systems lower recurring cost → more customers can be served.

6. Risks, Mitigation and Responsible Operations

Risk	How it could hurt the venture	Mitigation / decision rule
Resource risk	Target contains less accessible water than expected	Prospect first; independent data review; do not approve mining until resource confidence crosses a defined threshold
Technical risk	Anchoring, excavation, dust or thermal processing fails	Ground testing in representative environments; redundant contact points; sealed process path; pilot batch before scale-up
Transport risk	Trajectory or propulsion performance makes delivery too costly	Compare multiple targets; use mission-specific trajectory analysis; preserve schedule and propellant margin
Demand risk	No customer pays enough for delivered water	Secure customer interviews and conditional offtake agreements before full mission commitment
Schedule / finance risk	Long development cycles and cost overruns exhaust capital	Stage-gated financing; fixed test objectives; modular procurement; contingency reserve
Regulatory / governance risk	Authorization, liability, spectrum or resource-rights uncertainty	Obtain legal review in relevant jurisdictions; coordinate with responsible authorities; maintain transparent mission and safety plans
Environmental / debris risk	Operations create uncontrolled fragments or contaminate a site	Contained excavation; low-disturbance operations; debris tracking; end-of-life and disposal plan

Responsible operating principles

- Avoid uncontrolled release of excavated dust or fragments; contain material during extraction and transfer.
- Protect spacecraft and other operators through collision avoidance, conjunction monitoring and clear communications.
- Plan safe end-of-life disposal and avoid leaving non-functional equipment in hazardous or operationally sensitive locations.
- Document resource estimates, measurement uncertainty and delivery quality honestly; do not market unverified quantities.

A credible company is not one that says nothing can go wrong. It is one that knows what can go wrong, measures it early and has explicit stop/go rules.

7. Roadmap, KPIs and Team Execution

A staged roadmap prevents the venture from spending industrial-scale capital before the key unknowns are resolved.

Stage	Main work	Exit evidence
0. Feasibility	Target shortlist, trajectory screening, customer discovery, preliminary systems design	One target with plausible resource, access and customer case
1. Prospecting	Remote sensing and close-range characterization	Measured composition, terrain, spin and accessible deposit model
2. Technology demo	Rendezvous, anchoring, contained excavation and thermal extraction tests	Stable contact and repeatable small-batch recovery
3. Delivery demo	Purify, store, transfer and verify water at a depot or customer node	Accepted delivery with measured mass, purity and losses
4. Pilot service	Repeat deliveries with a contracted anchor customer	Reliable operations and a defensible delivered-cost model
5. Scale	Increase throughput; add modules, tugs and customer locations	Recurring demand and positive unit economics under conservative assumptions

Key performance indicators (KPIs)

- **Resource confidence:** estimated water abundance, accessible fraction and uncertainty range.
- **Extraction performance:** kilograms recovered per operating hour and recovery efficiency.
- **Energy intensity:** kilowatt-hours consumed per kilogram of accepted water.
- **System reliability:** uptime, fault frequency, mean time to recover and module replacement rate.
- **Delivery performance:** mass delivered, purity, loss rate, schedule reliability and customer acceptance.
- **Economics:** total mission cost, recurring cost per delivered kilogram, contracted demand and contribution margin.

Team execution priorities

The team should divide work into four tightly connected streams: (1) mission and orbital architecture, (2) mining and thermal processing, (3) cost and sensitivity model, and (4) customer and risk validation. One integrated systems owner must ensure that every subsystem assumption is reflected in mass, power, schedule and cost. For the pitch, each team member should be able to explain the complete end-to-end flow, not only their own slide.

8. Conclusion: The First Profitable Run

The Stack Avengers proposes a practical path toward asteroid-resource utilization by focusing on a product that has a clear in-space use: water. The system is designed around a reusable tug, autonomous prospecting, adaptable anchoring, contained excavation, thermal extraction, purification, standardized storage and delivery to a customer depot.

The strongest part of the proposal is its economic architecture. We do not assume that any asteroid is profitable simply because it contains valuable material. We select a target based on the entire delivered-cost chain, validate the resource before mining, sell where the resource has utility, and scale only after customers accept a measured delivery.

The first mission should be judged by four outcomes: (1) the target is characterized accurately, (2) a small quantity of water is extracted safely, (3) the water is stored and delivered with verified quality, and (4) the mission produces real cost and reliability data. These results turn a speculative concept into an investable engineering programme.

Han Solo's challenge is to prove that the mine can pay for itself. Our answer is not a giant mining ship on day one. It is a disciplined sequence of proof, delivery and reuse—one that learns before it scales.

The Stack Avengers

Team Lead: Yashica Singla
Team: The Stack Avengers
Problem Statement: STAC WARS — PS 4, Mining the Asteroid Belt
Core concept: Water-first, modular asteroid mining with reusable logistics and in-space customers.

Appendix A — Assumptions and limitations

- The report treats the year 2055 setting as a conceptual design horizon, not a claim that the proposed system is flight-ready today.
- No specific asteroid, orbit, launch vehicle, mission date, payload mass, extraction rate, power level, customer contract or selling price is supplied in the PS. These are intentionally left as items for engineering analysis and market validation.
- Solar-electric propulsion, autonomous navigation, thermal processing, solar arrays, storage tanks and robotic mechanisms are established technology families; their integration for this mission and the proposed force-balanced mining pod require substantial development and qualification.
- Water splitting into hydrogen and oxygen requires additional power, electrolysis equipment and cryogenic storage/handling. It is a future product option, not part of the minimum viable first delivery.
- This is a concept report and not a flight design, safety case, legal opinion or investment forecast.

Appendix B — PS 4 coverage checklist

PS requirement	Where addressed
Reach, land / operate around asteroid	Sections 2.1–2.3
Extract and process/refine material	Sections 2.1 and 3
Return to Earth / Moon / use in space	Sections 1 and 5.3
Energy source	Section 2.2
Communication, navigation, autonomy, failures	Sections 2.3–2.4
Engineering bottleneck and breakthrough	Section 3
Three largest cost drivers and cost reductions	Section 4
Conventional approach comparison	Section 4.2

Resource, customers, location, revenue model	Section 5
At least two business risks and mitigation	Section 6

Thank you

THE STACK AVENGERS | May the Code Be With You